

Adityavardhan Jain

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Professional Summary

Apps Engineering Analyst at Google building agentic SQL-generation workflows with deterministic validation (LLMs + SQLGlot). Prior Digital Analyst and Applied ML Enginner in computer vision (YOLO pothole detection, 95% mAP) and EEG signal classification. Published in the ACM Digital Library on handwritten medical text recognition; Best Paper at ACET-25.

Work Experience

Data Analytics Apprentice

Mar 2026 – Present

Google

Hyderabad, India

- Working in the **gTech Ads CX** team in an engineering-analyst capacity as an Apps Specialist, applying data analysis and statistical methods to investigate product and advertising performance issues.
- Developed internal automation workflows for **SQL generation and deterministic validation** using Agentic AI, SQLGlot, cloud computing, and statistical methods achieving **0.95** determinism score through metamorphic fuzzing.
- Analyzing campaign metrics, policies, and product configurations to identify opportunities across targeting, bidding strategies, account structure, and **ROAS**.
- Investigating and resolving engineering issues within the **Apps Advertisement** product area through data-driven root-cause analysis and product-level investigation.

Digital Analyst

Dec 2024 – Mar 2026

Tatvic Analytics

Ahmedabad, India

- Analyzed multi-channel datasets using **GA4, BigQuery, and SQL**, identifying conversion gaps and informing data-driven strategies that improved CVR by **4–5%**.
- Built automated analytical workflows using **n8n and Agentspace**, reducing manual reporting effort by **60%**.
- Performed **root-cause analysis** across client reporting pipelines, reducing data discrepancies by **80–90%**.
- Delivered **10+ advanced analyses** spanning attribution, cohort, and funnel analysis using SQL and statistical methods.
- Improved data pipelines and analytical processes to reduce latency by 12% and establish scalable, reproducible analysis workflows.

Computer Vision and Data Engineer

Jul 2024 – Dec 2024

Codenscious Technologies

Indore, India

- Built **Legolas**, a YOLOv11 based pothole detection system achieving **95% mAP**, with geospatial tagging and dashboards for road-condition monitoring.
- Designed and optimized a computer-vision pipeline for 50k+ images integrating **IoT, deep learning, geospatial data, and analytics**.
- Conducted research on **SLAM and ROS2 simulations**, evaluating model accuracy and applicability for robotics-oriented applications.

Data Science Research Intern

Jul 2023 – Jul 2024

Center of Advanced Research and Engineering, IIST

Indore, India

- Conducted research on **Brain-Computer Interfaces**, studying EEG signals to investigate cognitive effects associated with voluntary actions.
- Applied **deep learning, signal processing, statistical analysis, and data visualization** to EEG research and experimentation.

Selected Projects

Wiki Crawl: Interactive Knowledge Graph & Web Crawler

Sep 2026

- Built an interactive Wikipedia crawling and visualization system that explores relationships between pages and represents discovered knowledge as an interactive graph.
- Designed the crawling and graph traversal workflow with a focus on **efficient exploration, state management, caching, and scalable rendering** of interconnected information.
- Developed a browser-based interface for exploring dynamically discovered Wikipedia relationships and navigating connected knowledge spaces.

NeuroTetris: BCI-Controlled Gaming System

Aug 2023 – Jun 2024

- Built a BCI-based gaming system using **EEG signals**, applying machine learning and statistical modeling to classify neural patterns with **80% accuracy** of win/loss.
- Developed an EEG preprocessing pipeline involving **signal filtering, feature extraction, exploratory analysis, and noise/variability analysis**.
- Implemented real-time control mapping in Python to translate classified neural states into game interactions.

Research Publication

A Method for Doctor’s Handwritten Text Recognition by Utilizing Deep Learning and Ambiguity Handling Jun 2026

- Published research in the **ACM Digital Library** on deep-learning approaches for handwritten medical text recognition and document understanding.
- Investigated **ambiguity-aware recognition** methods for interpreting complex and context-dependent clinical handwriting.
- Applied techniques from **computer vision, OCR, and NLP** to healthcare document intelligence.
- Co-authored and presented research on automated extraction of information from handwritten medical records.

Technical Skills

Programming: Python, SQL, C++, R, Bash, Data Structures & Algorithms

Machine Learning: PyTorch, TensorFlow, Scikit-learn, Supervised & Unsupervised Learning, Feature Engineering, Statistical Modeling, Model Evaluation, Computer Vision, NLP

Generative AI: LLMs, Transformers, Generative AI, Agentic AI, RAG, Prompt Engineering, Embeddings, Vector Databases, LLM Evaluation

Data Engineering: Pandas, NumPy, PySpark, Hadoop, ETL, Data Pipelines, BigQuery, Data Processing

ML Engineering: Model Deployment, MLOps, REST APIs, Flask, FastAPI, Docker, CI/CD, Git, Linux

Cloud & Platforms: GCP, Azure Databricks, Google Agentspace, Google AI Studio

Specialized: OCR, Image Processing, EEG Signal Processing, Brain-Computer Interfaces, ROS2, SLAM

Education

Indore Institute of Science and Technology
B.Tech. in Artificial Intelligence and Machine Learning (CGPA: 8.03/10)

May 2025
Indore, Madhya Pradesh, India

Awards & Leadership

Winner, Smart India Hackathon 2023 – Software Edition — Ministry of Defence Computer Vision problem statement of Robust Human Target Detection and Acquisition.

Best Paper Award — “Neural Signatures of Eye Blinking: EEG Analysis for Inferencing Complex Cognitive Processes.” at the International Conference on Advances in Cutting-Edge Engineering and Technology (ACET-25)

Google Developer Student Clubs Lead (2023–24) and **CodeChef IIST Chapter Lead (2024–25)**.

Certifications: Google Advanced Data Analytics Professional, Google Prompt Engineering Professional, Google AI Professional, Harvard/edX Python for Data Science, Harvard/edX Data Science: Inference and Modelling

Spot Award and Recognition – Consulting critical clients at Tatvic